Homework 2

See Canvas for due date and time

Please be aware of our *AI policy* per the course syllabus.

1 Implementing MCTS

Read and modify the provided hw2.ipynb file to implement the following methods therein. It is recommended to read lecture2a.pdf on Canvas or an equivalent as a reference for your implementation.

(a) [1 point] `tree_policy` in class `MCTS` when transition (s, a, s') is already in the tree.

(b) [1 point] `tree_policy` in class `MCTS` when transition (s, a, s') is unseen in the tree.

(c) [2 points] `best_action` in class `MCTS`.

(d) [2 points] `search` in class `MCTS`.

(e) [2 points] `select_action` in class `UCT`.

The original hw2.ipynb file provides a test at the end, which only serves as a reference for grading. After implementing all methods above, run the test and answer the following question:

(f) [2 points] Can you briefly summarize your test result and interpret it?

2 Analysis of MAB Algorithms

Read lecture3a.pdf on Canvas carefully.

(a) [8 points] Answer the questions tagged as (1a)-(1d) in lecture3a.pdf, 2 points each. The questions are *underlined and in italics*.

3 Policy Optimization, Implmenting PPO

(a) [3 points] Read lecture3c.pdf on Canvas and justify the derivations tagged as **2a-2c** therein, 1 points each.

(b) [9 points] Read hw3.ipynb that implements a simplified version of PPO. Fill in your implementation of the unfinished methods therein. Details can be found in hw3.ipynb.